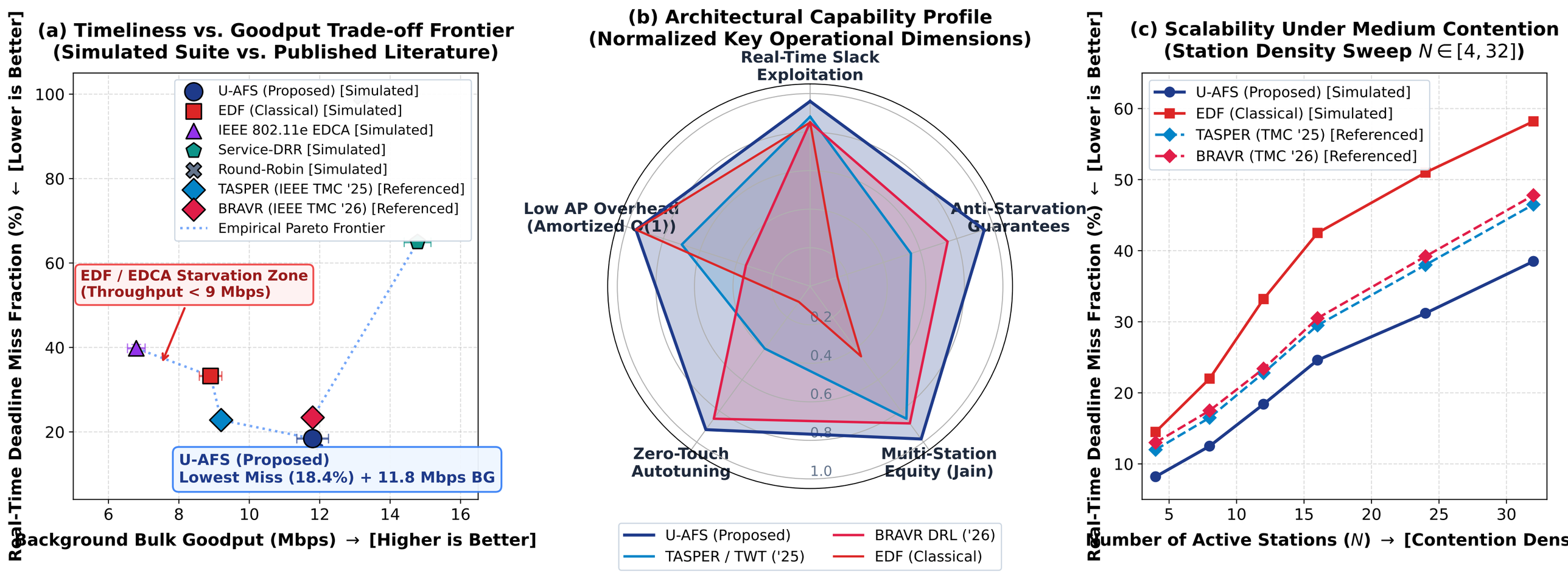
(a) Timeliness vs. Goodput Trade-off Frontier
(Simulated Suite vs. Published Literature)
U-AFS (Proposed) [Simulated]
EDF (Classical) [Simulated]
IEEE 802.11e EDCA [Simulated]
Service-DRR [Simulated]
Round-Robin [Simulated]
TASPER (IEEE TMC '25) [Referenced]
BRAVR (IEEE TMC '26) [Referenced]
Empirical Pareto Frontier
EDF / EDCA Starvation Zone
(Throughput < 9 Mbps)
U-AFS (Proposed)
Lowest Miss (18.4%) + 11.8 Mbps BG
Real-Time Deadline Miss Fraction (%) ← [Lower is Better]
Background Bulk Goodput (Mbps) → [Higher is Better]
(b) Architectural Capability Profile
(Normalized Key Operational Dimensions)
Real-Time Slack
Exploitation
Anti-Starvation
Guarantees
Multi-Station
Equity (Jain)
Zero-Touch
Autotuning
Low AP Overhead
(Amortized O(1))
U-AFS (Proposed)
TASPER / TWT ('25)
BRAVR DRL ('26)
EDF (Classical)
(c) Scalability Under Medium Contention
(Station Density Sweep $N \in [4, 32]$)
U-AFS (Proposed) [Simulated]
EDF (Classical) [Simulated]
TASPER (TMC '25) [Referenced]
BRAVR (TMC '26) [Referenced]
Real-Time Deadline Miss Fraction (%) ← [Lower is Better]
Number of Active Stations (N) → [Contention Dens